

Hồ Lam Sơn

Name also written as: Lam-Son Ho, Son Lam Ho
Languages: Vietnamese (native), English (fluent)

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PROFILE

Data scientist and Math PhD (University of Maryland) with nine years of experience leading machine learning projects in industry. Teaching experience in university courses in Statistics, Linear Algebra, Calculus, as well as Foundation of Machine Learning workshops at VietAI.

EDUCATION

Ph.D., Mathematics August 2014

University of Maryland, College Park, MD

Advisor: Dr. William M. Goldman

B.A. with Honors, Mathematics and Computer Science (Double Major) June 2008

Williams College, Williamstown, MA

Advisor: Dr. Thomas Garrity

PROFESSIONAL POSITIONS

Lead Data Scientist Sep 2017 – Sep 2026

Urbanmetry Sdn Bhd, Malaysia

Designed, implemented, and refined all stages of several machine learning projects, including:

- **Home price prediction:** a model used by banks to assess loan applications and by our sister company UrbanNX in its investment decision process.
- **Bus ridership prediction:** a model used by major bus operators in Kuala Lumpur and Penang to plan changes to their bus networks.
- **Internal models:** A text classifier using Convolution Neural Nets, an image segmentation model that detects houses in satellite imagery of rural areas.

Other notable software projects:

- A high-performance exchange order book in Go for trading fractionalized properties.

Provided data science and technical training for selected team members.

Teaching Team Member (Hanoi) 2018 – 2019

VietAI, Vietnam

Taught sessions for “Foundations of Machine Learning” course at VietAI, a non-profit organization building a community of AI practitioners and experts in Vietnam. Course topics: Intro to Neural Networks, Back-propagation algorithm and modifications, loss functions, regularization methods.

Postdoctoral Researcher, Mathematics Aug 2015 – Aug 2017

Université de Sherbrooke, Quebec, Canada

Advisor: Dr. Virginie Charette

Postdoctoral Researcher, Mathematics Sep 2014 – Mar 2015

University of Luxembourg, Luxembourg

Advisor: Dr. Jean-Marc Schlenker

UNIVERSITY TEACHING EXPERIENCE

Lecturer

Planned lessons and assignments, lectured 3 hours per week per section (5 hours per week in summer terms), held office hours, and graded coursework.

STAT 100	Elementary Statistics and Probability · UMD	Spring 2014
MATH 221	Elementary Calculus II · UMD	Summer 2013
MATH 461	Linear Algebra for Scientists and Engineers · UMD	Summer 2011
STAT 100	Elementary Statistics and Probability · UMD	Fall 2009

Teaching Assistant

Helped plan assignments, taught or led discussion sessions 1.5 hours per week, and graded coursework.

Analyse 1	Real Analysis 1 · Luxembourg	Fall 2014
MATH 140	Calculus I · UMD	Fall 2010
MATH 220	Elementary Calculus I · UMD	Spring 2010

Grader

Graded coursework and answered students' questions.

MATH 432	Introduction to Topology · UMD	Fall 2012
MATH 431	Geometry for Computer Graphics · UMD	Fall 2012
MATH 341	Linear Algebra and Differential Equations · UMD	Spring 2012

TECHNICAL SKILLS

- **Data science:** Python (NumPy, pandas, scikit-learn, SciPy, XGBoost, Tensorflow) and SQL (PostgreSQL)
- **Other languages and frameworks:** Go, Ruby on Rails, Clojure
- AI-assisted software development with Claude Code

RESEARCH & PUBLICATIONS

- Chapter 8 of *Eighteen Essays in Non-Euclidean Geometry*. [EMS IRMA Lectures in Mathematics and Theoretical Physics](#), 2019.
- “Deformations of fundamental group representations and earthquakes on $SO(n, 1)$ surface groups.” *Geometriae Dedicata*, 2017. [doi:10.1007/s10711-017-0226-0](https://doi.org/10.1007/s10711-017-0226-0)
- “On conformally flat circle bundles over surfaces.” Ph.D. thesis, University of Maryland, 2014. Preprint: [arXiv:1412.5824](https://arxiv.org/abs/1412.5824)

REFERENCES

Available on request.